

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 October 2019 Worked Solutions

October 2019 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 October 2019

October 2019 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Sequences & Series

1. A curve C has equation $y = 2x^2(x - 5)$

(a) Find, using calculus, the x coordinates of the stationary points of C .

(4)

(b) Hence find the values of x for which y is increasing.

(2)

(Total 6 marks)

Worked Solution - Question 1

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 2

Sequences & Series

2. The adult population of a town at the start of 2019 is 25 000

A model predicts that the adult population will increase by 2% each year, so that the number of adults in the population at the start of each year following 2019 will form a geometric sequence.

- (a) Find, according to the model, the adult population of the town at the start of 2032 (3)

It is also modelled that every member of the adult population gives £5 to local charity at the start of each year.

- (b) Find, according to these models, the total amount of money that would be given to local charity by the adult population of the town from 2019 to 2032 inclusive. Give your answer to the nearest £1 000 (3)

(Total 6 marks)

Worked Solution - Question 2

Topic group

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: - - or () 14 125000 1

Final answer

- - or () 14 125000 1

Question 3

Sequences & Series

3. (a) Find the first 4 terms, in ascending powers of x , in the binomial expansion of

$$\left(1 + \frac{x}{4}\right)^{12}$$

giving each coefficient in its simplest form.

(3)

- (b) Find the term independent of x in the expansion of

$$\left(\frac{x^2 + 8}{x^5}\right)\left(1 + \frac{x}{4}\right)^{12}$$

(3)

(Total 6 marks)

Worked Solution - Question 3

Topic group

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: $55\ 99\ 77\ 16\ 16\ 8\ +\ =$

Final answer $55\ 99\ 77\ 16\ 16\ 8\ +\ =$

Question 4

Sequences & Series

4. $f(x) = (x - 3)(3x^2 + x + a) - 35$ where a is a constant

(a) State the remainder when $f(x)$ is divided by $(x - 3)$.

(1)

Given $(3x - 2)$ is a factor of $f(x)$,

(b) show that $a = -17$

(2)

(c) Using algebra and showing each step of your working, fully factorise $f(x)$.

(5)

(Total 8 marks)

Worked Solution - Question 4

Topic group

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: - + -

Final answer

- + -

Question 5

Sequences & Series

5. (a) Given $0 < a < 1$, sketch the curve with equation

$$y = a^x$$

showing the coordinates of the point at which the curve crosses the y-axis.

(2)

x	2	2.5	3	3.5	4
y	4.25	6.427	9.125	12.34	16.06

The table above shows corresponding values of x and y for $y = x^2 + \left(\frac{1}{2}\right)^x$

The values of y are given to 4 significant figures as appropriate.

Using the trapezium rule with all the values of y in the given table,

- (b) obtain an estimate for $\int_2^4 \left(x^2 + \left(\frac{1}{2}\right)^x \right) dx$

(3)

Using your answer to part (b) and making your method clear, estimate

- (c) $\int_2^4 \left(x(x-3) + \left(\frac{1}{2}\right)^x \right) dx$

(2)

(Total 7 marks)

Worked Solution - Question 5

Topic group

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: + - = - =

Final answer

+ - = - =

Question 6

Sequences & Series

6.

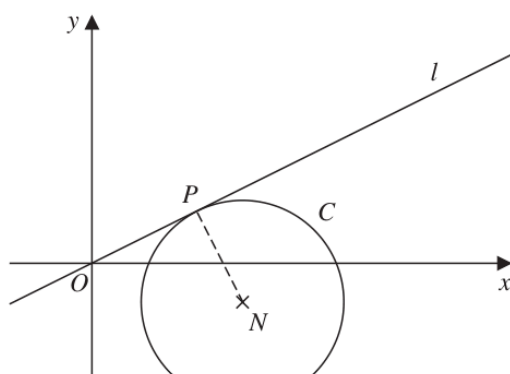


Figure 1

Figure 1 shows a sketch of a circle C with centre $N(4, -1)$.

The line l with equation $y = \frac{1}{2}x$ is a tangent to C at the point P .

Find

(a) the equation of line PN in the form $y = mx + c$, where m and c are constants, (2)

(b) the equation of C . (5)

(Total 7 marks)

Worked Solution - Question 6

Topic group

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: $- + -- = oe () 2 2 36 4 5 x$
 $y - + + = oe$

Final answer

$- + -- = oe () 2 2 36 4 5 x y - + + = oe$

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7 marks

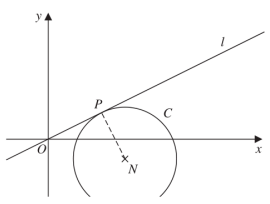
Question 7

Laws of Logarithms

The table above shows corresponding values of x and y for $y = x^2 + \left(\frac{1}{2}\right)^x$. The values of y are given to 4 significant figures as appropriate. Using the trapezium rule with all the values of y in the given table, (b) obtain an estimate for $\int_2^4 \left(x^2 + \left(\frac{1}{2}\right)^x\right) dx$ Using your answer to part (b) and making your method clear, estimate (c) $\int_2^4 \left(x(x-3) + \left(\frac{1}{2}\right)^x\right) dx$	(3) (2)
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Question 5 continued	blank
Question 5 continued	blank

Question 5 continued	blank
(Total 7 marks)	

6.  Figure 1 Figure 1 shows a sketch of a circle C with centre $N(4, -1)$. The line l with equation $y = \frac{1}{2}x$ is a tangent to C at the point P. Find (a) the equation of line PN in the form $y = mx + c$, where m and c are constants, (b) the equation of C.	blank (2) (5)
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Question 6 continued	blank
Question 6 continued	blank

Question 6 continued	blank
(Total 7 marks)	

7. Given $\log_b b = k$, find, in simplest form in terms of k, (i) $\log_b \left(\frac{\sqrt{a}}{b}\right)$ (ii) $\frac{\log_a a^2 b}{\log_a b^3}$ (iii) $\sum_{n=1}^{50} (k + \log_a b^n)$	(2) (2) (3)
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(Total 7 marks)	
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Worked Solution - Question 7

Topic group

1. Combine the logarithms

Use the power, product, and quotient laws to rewrite the equation as a single logarithmic or exponential statement.

2. Solve with domain checks

Solve the resulting equation and reject any value that does not satisfy the original logarithms.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

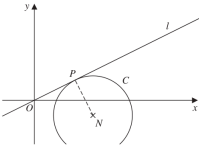
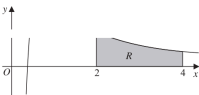
The required value or expression is obtained by the final simplification.

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9 marks

Question 8

Laws of Logarithms

Using your answer to part (b) and making your method clear, estimate	
(c) $\int_2^4 \left(x(x-3) + \left(\frac{1}{x} \right) \right) dx$	(2)
Question 5 continued	blank
Question 5 continued	blank
Question 5 continued	blank
(Total 7 marks)	
6.	blank
 Figure 1 Figure 1 shows a sketch of a circle C with centre $M(4, -1)$. The line l with equation $y = \frac{1}{2}x$ is a tangent to C at the point P. Find (a) the equation of line PN in the form $y = mx + c$, where m and c are constants, (2) (b) the equation of C. (5)	
Question 6 continued	blank
Question 6 continued	blank
Question 6 continued	blank
(Total 7 marks)	
7. Given $\log_a b = k$, find, in simplest form in terms of k ,	blank
(i) $\log_a \left(\frac{\sqrt{a}}{b} \right)$	(2)
(ii) $\frac{\log_a a^2 b}{\log_a b^2}$	(2)
(iii) $\sum_{r=1}^{20} (k + \log_a b^r)$	(3)
Question 7 continued	blank
(Total 7 marks)	
8. Solutions relying on calculator technology are not acceptable in this question.	blank
(i)	
 Figure 2 Figure 2 shows a sketch of part of a curve with equation $y = \frac{8\sqrt{x}-5}{2x^2} \quad x > 0$ The region R, shown shaded in Figure 2, is bounded by the curve, the line with equation $x = 2$, the x-axis and the line with equation $x = 4$. Find the exact area of R. (5) (ii) Find the value of the constant k such that $\int_2^4 \left(\frac{1}{2}x^2 + k \right) dx = 55$ (4)	
(Total 9 marks)	

Worked Solution - Question 8

Topic group

1. Combine the logarithms

Use the power, product, and quotient laws to rewrite the equation as a single logarithmic or exponential statement.

2. Solve with domain checks

Solve the resulting equation and reject any value that does not satisfy the original logarithms.

3. State the final result

Write the answer in the form requested by the question: $- + = - + -- - = - = 29\ 18$
 $k =$

Final answer

$- + = - + -- - = - = 29\ 18\ k =$

Question 9

Trigonometric Equations

9. Solutions based entirely on graphical or numerical methods are not acceptable in this question.

- (i) Solve, for $0 \leq \theta < 180^\circ$, the equation

$$3 \sin(2\theta - 10^\circ) = 1$$

giving your answers to one decimal place.

(4)

- (ii) The first three terms of an arithmetic sequence are

$$\sin \alpha, \frac{1}{\tan \alpha} \text{ and } 2 \sin \alpha$$

where α is a constant.

- (a) Show that $2 \cos \alpha = 3 \sin^2 \alpha$

(3)

Given that $\pi < \alpha < 2\pi$,

- (b) find, showing all working, the value of α to 3 decimal places.

(5)

(Total 12 marks)

Worked Solution - Question 9

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: radians

Final answer

radians

Question 10

Sequences & Series

10. The curve C has equation

$$y = ax^3 - 3x^2 + 3x + b$$

where a and b are constants.

Given that

- the point $(2, 5)$ lies on C
- the gradient of the curve at $(2, 5)$ is 7

(a) find the value of a and the value of b .

(4)

(b) Prove that C has no turning points.

(3)

Q10

(Total 7 marks)

Worked Solution - Question 10

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: - hence there are no turning points oe *

Final answer

- hence there are no turning points oe *